

## Demonstration of Proposed Features

|                      |                                       |
|----------------------|---------------------------------------|
| <b>Date</b>          | 04 July 2026                          |
| <b>Team ID</b>       |                                       |
| <b>Project Name</b>  | A Comprehensive Measure of Well-Being |
| <b>Maximum Marks</b> | 1 Mark                                |

## Demonstration of Proposed Features

This document summarizes the implemented features of the **Human Development Index (HDI) Prediction System** and their completion status. It serves as evidence that the proposed functionalities were successfully developed, tested, and demonstrated through the Flask web application.

| S.No | Feature Name                   | Description                                                                                                                                      | Status<br>(Implemented /<br>Partial /<br>Pending) | Demonstrated<br>(Yes / No) | Remarks                                  |
|------|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|----------------------------|------------------------------------------|
| 1    | Session-Based Authentication   | Implemented a secure login system using session-based authentication to restrict unauthorized access to the application dashboard.               | Implemented                                       | Yes                        | Successfully completed                   |
| 2    | Country Metric Prefill         | Automatically fills Life Expectancy, Expected Years of Schooling, Mean Years of Schooling, and GNI per Capita values when a country is selected. | Implemented                                       | Yes                        | Dynamic country data loaded successfully |
| 3    | Interactive Simulation Sandbox | Developed interactive sliders and synchronized input fields that allow users to modify development indicators in real time.                      | Implemented                                       | Yes                        | User-friendly interface implemented      |
| 4    | HDI Prediction & Suggestions   | Predicts the Human Development Index score and category using the trained Machine Learning                                                       | Implemented                                       | Yes                        | Accurate prediction results generated    |

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|------|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|----------------------------|----------------------------------|
|      |                                 | model and provides personalized policy suggestions.                                                                                      |                                                   |                            |                                  |
| 5    | Prediction History & CSV Export | Stores previous prediction records using Local Storage and allows users to export prediction history as a CSV file.                      | Implemented                                       | Yes                        | Successfully demonstrated        |
| 6    | Flask Web Application           | Developed a responsive Flask web application integrating the trained machine learning model, frontend interface, and prediction modules. | Implemented                                       | Yes                        | Successfully deployed and tested |

### Feature Implementation Summary

| Description                     | Count |
|---------------------------------|-------|
| Total Features Proposed         | 6     |
| Total Features Implemented      | 6     |
| Total Features Demonstrated     | 6     |
| Overall Implementation Rate (%) | 100%  |